Riccardo Michielan

riccardo.michielan@gssi.it

Website $\mathcal{R}^G$ ResearchGate

ORCID: 0000-0003-4642-8507



Research

My research lies at the intersection of probability, random graphs, and complex networks, with a focus on scale-free networks, network geometry, subgraph counts, motifs, and random graph algorithms. Current interests include planted structure recovery, short-time random walks on networks, large deviations for scale-free graph observables, dynamic community detection, and temporal graph connectivity.

› POSITIONS

Feb 2027 - Jan 2029 — **MSCA Postdoc fellow | University of Padua (not signed yet)**
Position: *Incarico postdoc ai sensi dell'art. 22-bis della Legge 240/2010.*
Project: *PAiNe - Planted anomalies in networks*
Research unit: *Probability group, Department of Mathematics*
Supervisor: *Prof. Alessandra Bianchi*

Feb 2025 – Jan 2027 — **Postdoc researcher | Gran Sasso Science Institute**
Position: *Assegno di ricerca ai sensi dell'art. 22 della Legge n. 240/2010.*
Project: *GAMING - Graph algorithms and mining for green agents*
Research unit: Computer Science
Supervisor: *Prof. Gianlorenzo D'Angelo*

Nov 2020 – Feb 2025 — **PhD | University of Twente**
Position: *Doctoral candidate (promovendus) under the CAO-NU*
Research unit: *Mathematics of Operation Research, EEMCS*
Supervisor: *Prof. Marc Uetz, Prof. Clara Stegehuis*

› GRANTS & FUNDING

Awarded MSCA Postdoctoral Fellowship 2025
Title: *PAiNe - Planted anomalies in networks*
Role: *Principal Investigator / Fellow*
Funding award: *€ 193643, Horizon Europe > Marie Skłodowska-Curie Actions*

Researcher on PNRR FAIR cascade project
Title: *GAMING - Graph algorithms and mining for green agents*
Role: *Postdoc researcher*
Project funding: *€ 200000, European Union > NextGenerationEU*

› VISITING

Apr 2023 – Jun 2023 — **Stockholm University, Department of Mathematics**
Visiting Prof. Maria Deijfen, Erasmus+ KA131 Staff mobility

Selected talks

› CONTRIBUTED TALKS

Jun 2026 — **Workshop on Modelling and Mining Networks (WAW)** – Toronto, Canada
Talk: *Planted clique recovery in random geometric graphs*
Event link

Jun 2026 — **Italian Meeting on Probability and Mathematical Statistics** – Palermo, Italy
Talk: *Homophily within and across groups*
Event link

Jul 2024 — **Saint-Flour 52nd Summer school in probability** – Saint-Flour, France
Talk: *Divide et impera: optimal subgraphs*
Event link

Mar 2024 — **Young European Probabilists (YEP)** – Eindhoven, Netherlands
Talk: *Detecting network geometry using global and local triangle-based statistics*
Event link

Jun 2022 — **Italian Meeting on Probability and Mathematical Statistics** – Bologna, Italy
Talk: *Influence of geometry in the structure of large scale-free networks*
Event link

Dec 2021 — **Conference of Complex Networks** – Madrid, Spain
Talk: *Cliques in geometric power-law random graphs: a phase transition for the influence of geometry*
Event link

Sep 2021 — **Randomness unleashed (GTD)** – Groningen, Netherlands
Talk: *Subgraphs in geometric network models*
Event link

› INVITED TALKS

Oct 2025 — **SINEQ Conference** – L'Aquila, Italy
Talk: *Connectivity and shortest paths of temporal random graphs*
Event link

Mar 2024 — **CX Talks** – Aalto, Finland
Talk: *Modeling networks through geometric random intersection graphs*
Event link

Mar 2023 — **iPOD seminars** – Leiden, Netherlands
Talk: *Distinguishing, detecting and identifying geometry in networks through "triangle-based" statistics*
Event link

Feb 2023 — **Special Guest Seminar Series** – Florence, Italy
Talk: *Is there geometry in real networks?*
Event link

› ORGANIZED SESSIONS

Jun 2026 — **Italian Meeting on Probability and Mathematical Statistics** – Palermo, Italy
Session: *Community structure of Complex Networks*
Event link

Teaching experience

› TEACHING & TUTORING

2022 – 2024 **Teacher Assistant for the course *Complex Networks***
Type: *Curricular course for the Master Applied Mathematics*
Credits: *5 ECTS (5 CFU)*
Institute: *EEMCS Faculty, University of Twente*
Total load: *approx. 80 hours/year*

2022 – 2024 **Tutor for the course *Calculus 1B***
Type: *Tutoring activity*
Institute: *EEMCS Faculty, University of Twente*
Total load: *approx. 60 hours/year*

2022 – 2023 **Tutor for the course *Calculus A***
Type: *Tutoring activity*
Institute: *EEMCS Faculty, University of Twente*
Total load: *56 hours*

2018 – 2020 **Tutor for the courses *Mathematical Analysis I* and *Linear Algebra***
Type: *Tutoring activity*
Institute: *Industrial Engineering, University of Padua*
Total load: *on average 90 hours/year*

› MENTORING

2022– 2023 **Supervisor for *Bachelor Thesis***
Title: *How well can personal network features be retrieved?*
Student: *Bo Arends*
Curriculum: *Applied Mathematics, University of Twente*

Education

Oct 2017 – Oct 2020 **MSc in Mathematics (cum laude) | University of Padua**
Thesis: *An algorithmic approach to the random spanning forests*
Advisor: *Prof. Alessandra Bianchi*

Oct 2013 – Sep 2017 **BSc in Mathematics | University of Padua**
Thesis: *Large deviations principle for RW in random environments*
Advisor: *Prof. Alessandra Bianchi*

› SKILLS

Languages	Italian (native); English; Spanish
Coding	Python (incl. *numpy*, *scipy*, *networkx*), Mathematica, Matlab.
Web	HTML, CSS, JavaScript
Design	Inkscape, Adobe Creative Cloud (incl. *Photoshop*)

Luogo e data:
Roma, August 12, 2026

Firma



Riccardo Michielan